SEPTEMBER
2026

Orifice Plate Sizing Tool

ENGINEERING VALIDATION REPORT BASED ON BS EN ISO 5167

PREPARED BY: GHODOUSI BOROUJENI, MAHSHAD

REVIEWED BY:TALESHI, SHAHPOUR

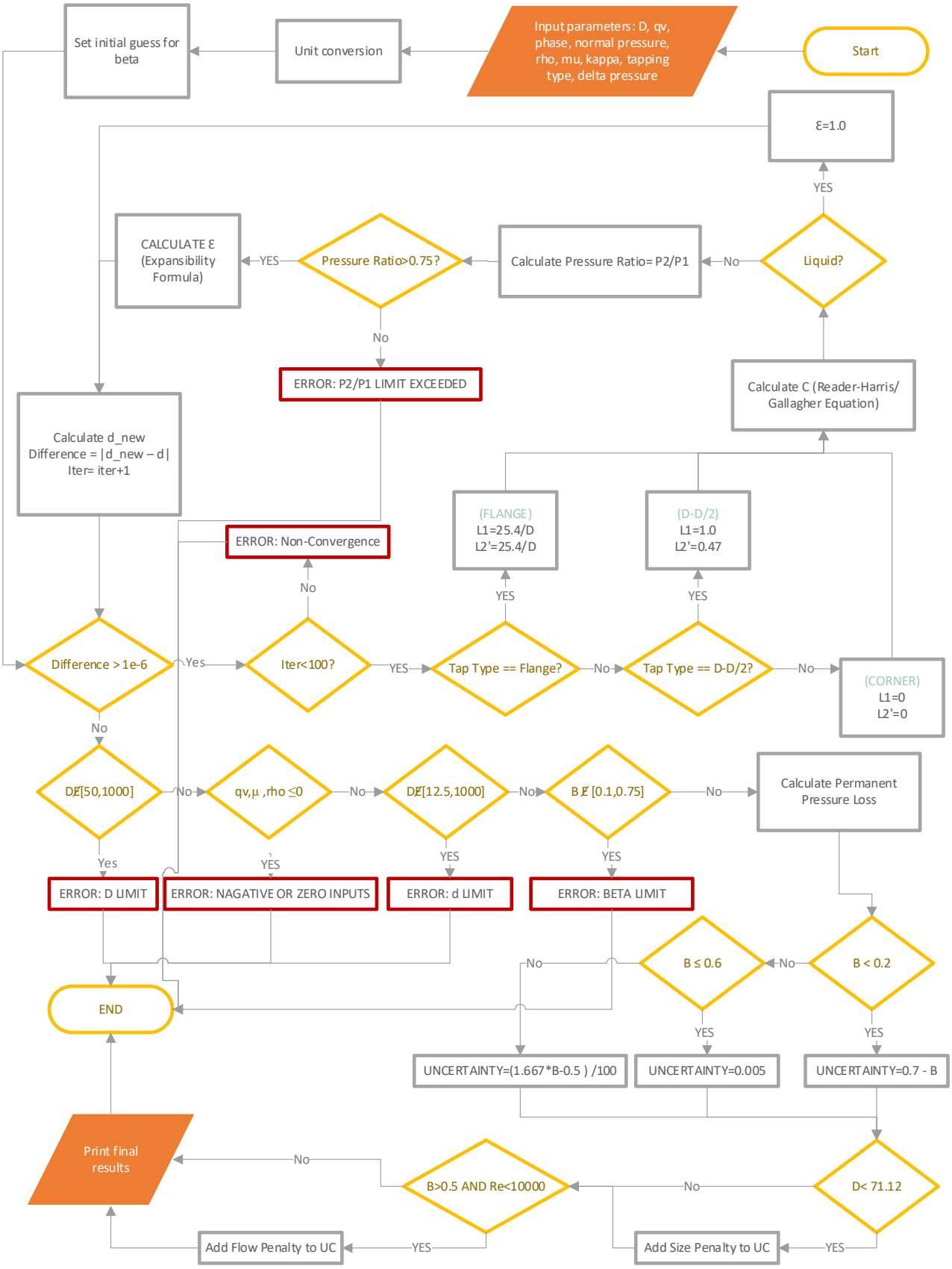
1. Purpose and Scope:

The purpose of this document is to present and validate a custom C++ computational tool designed for orifice plate sizing. The algorithm was developed in strict accordance with the BS EN ISO 5167-2 standard requirements. To verify the reliability of the software for industrial applications, the calculated outputs-specifically the orifice bore diameter, beta ratio, and permanent pressure loss- are cross-referenced against approved engineering datasheets for existing Plant Air and Plant Water services.

2. Computational Methodology:

Due to the non-linear interdependence between the orifice bore diameter (d), the pipe Reynolds number (Re), and the discharge coefficient (C), a direct analytical solution is not feasible. To resolve this, the software utilizes a successive substitution algorithm. The program initiates with an estimated beta ratio and iteratively recalculates the discharge coefficient (C) and expansibility factor (Ɛ). The numerical loop is configured to evaluate the error margin dynamically and terminates only when the difference between successive bore diameter calculations falls below a tolerance threshold of 10^{-6}. This approach ensures the final physical dimensions meet standard manufacturing tolerances while preventing infinite loop execution during edge-case inputs.

3. Algorithm Flowchart



4. Validation and Results

To verify the accuracy and reliability of the developed sizing algorithm, the computational outputs were cross-referenced against approved engineering datasheets for existing field instruments. The validation encompasses both compressible (Plant Air) and incompressible (Plant Water) fluid services.

4.1. Validation Case 1: Plant Air (Tag: 42-FE-4252)

Parameter	Symbol / Unit	Datasheet Value	C++ Algorithm Output	Deviation (%)
Orifice Bore Diameter	d (inch)	1.514	1.514	0%
Beta Ratio	Beta	0.4935	0.4936	0.02%
Permanent Pressure Loss	Pressure loss (mbar)	739.54 mbar	739.45 mbar	0.01%

NOTE: The original engineering datasheet and the raw computation terminal outputs for this validation case are provided in Appendix A and B ,respectively.

4.2. Validation Case 2: Plant Water (Tag: 42-FE-4272)

Parameter	Symbol / Unit	Datasheet Value	C++ Algorithm Output	Deviation (%)
Orifice Bore Diameter	d (inch)	1.395	1.396	0.07%
Beta Ratio	Beta	0.4549	0.4551	0.04%
Permanent Pressure Loss	Pressure loss (mbar)	96.84 mbar	96.83 mbar	0.01%

NOTE: The original engineering datasheet and the raw computation terminal outputs for this validation case are provided in Appendix A and B ,respectively.

5. Conclusion

The development and validation of the C++ orifice sizing algorithm demonstrate its high accuracy, robustness, and strict compliance with the BS EN ISO 5167 standard. By cross-referencing the computational outputs with approved industrial datasheets for both compressible (Plant Air) and incompressible (Plant Water) fluid services, the maximum observed deviation across all critical parameters-including bore diameter, beta ratio, and permanent pressure loss-was consistently below 0.1%.

This exceptional level of precision confirms that the underlying logic, particularly the successive substitution loop for the discharge coefficient and expansibility factor, is fundamentally sound. Furthermore, the successful integration of strict hydrodynamic and geometric constraints ensures that the tool autonomously filters out-of-bound inputs and maintains computational integrity. Additionally, the algorithm dynamically adapts to various pressure tapping arrangements-including Flange, Corner, and D and D/2 taps-by automatically adjusting the corresponding L_1 and L_2' coefficients, ensuring its versatility across diverse industrial piping configurations. Ultimately, the developed algorithm proves to be a highly reliable engineering tool, fully capable of supporting precise instrumentation sizing, verification, and quality control tasks.

6. Appendices

Document No.:		ER-142-FEA-INS-DS-011					Contract No.:	پ ن الف/73-02-1941			Sheet No.: 4 of 30
GENERAL	1	Tag Number						42-FE-4252			
	2	Service			Service Type			Plant Air to Unit 142			Continuous
	3	Min. Ambient Temp.			Max. Ambient Temp.		Unit	-15		41	°C
	4	Max.Relative Humidity					Unit	73.00			%
	5	P & ID No.			Unit No.			ER-142-FSD-DI-001		Interconecting	
	6	Line No.			Piping Class	Line Thickness		3"-PA-1423305-1BD1-NI		1BD1	S-STD
PROCESS CONDITIONS	7	Fluid			Phase			Plant Air		Gas	
	8	Flow:	Minimum	Normal	Maximum		Unit	-	-	3300.00	kg/h
	9	Temperature	Minimum	Normal	Maximum		Unit	-	40.0	-	°C
	10	Pressure:	Minimum	Normal	Maximum		Unit	-	6.9	-	bar g
	11	Base Temperature			Design Temperature		Unit	-		65	°C
	12	Base Pressure			Design Pressure		Unit	-		10.3	bar g
	13	Full Scale Flow					Unit	3300			kg/h
	14	Maximum allowable pressure loss					Unit	1.0			bar
	15	Operating Density			Density @ Base		Unit	8.618		-	kg/m3
	16	Compressibility Factor (Z)						0.9969			
	17	Molecular Weight			Cp / Cv			28.95		1.421	
	18	Vapor Pressure Pv					Unit	-			bar g
	19	Maximum Velocity					Unit	23.33			m/s
	20	Vaccum Possibility			NACE Requirement			No		No	
	21	Operating Viscosity					Unit	1.99e-002			cp
	22	Corrosive Service			Cold Service			No		No	
ORIFICE PLATE	23	Orifice Plate Type						square-edged concentric			
	24	Standard						ISO 5167			
	25	Material						SS 316			
	26	Plate Thickness						3 mm			
	27	Pipe inside diameter (20°C)						3.068 in			
	28	Pipe Reynolds No. @ Operating Flow						752450.25969			
	29	Beta Ratio @ 20°C						0.49350586914			
	30	Bore Dia @ 20°C						1.514 in			
	31	Drain or Vent Hole Size			Hole Position			N/A (Note *)		N/A (Note *)	
	32	Residual pressure loss						739.54			mbar
	33	Sound pressure level						62.90			dB(A)
	34	Ring Material & Type						N/A			
	35	Taps Type	Taps Size		Taps Orientation			Flange	1/2" NPT	45° upward Horizontal	
	36	Flange Size			Flange Type & Rating			3"		RF & 300#	
	37	Flange Material			Gasket Material			A105		SPW,SS 316L Winding & Inner Ring,CS Outer Ring,Graphite Filler	
	38	Permanent Pressure Loss						739.54 mbar			
	39	Spacer Ring						N/A			
METER	40	Transmitter Spec.						Model: 2051C D 3 A 0 2 A 1 A S5 B4 I1 M5 Q4 Q8			
	41	Differential Pressure @ Full scale						1000 mbar			
	42	Meter Max Flow						0 ~ 3300 kg/m3			
	43										
	44										
PURCHASE	45	Manufacturer						ARAMAK / IRAN			
	46	Model						Orifice Plate Primary :OFT-RF1-80-I1-A4-A2-I4-DH3-HA1-0-01-C0-DW-SG2 - GS-T02			
	47	Requisition Number			Item Number			-		-	
	48	Serial Number						N/A			

Figure A.1: Approved engineering datasheet for Plant Air Tag 42-FE-4252

Document No.:		ER-142-FEA-INS-DS-011				Contract No.:	ب ن الف/73-02-1941			Sheet No.: 7 of 30
GENERAL	1	Tag Number					42-FE-4272			
	2	Service		Service Type			Plant water to Unit 142			Continuous
	3	Min. Ambient Temp.		Max. Ambient Temp.		Unit	-15	41		°C
	4	Max.Relative Humidity				Unit	73.00			%
	5	P & ID No.		Unit No.			ER-142-FSD-DI-004	Interconecting		
	6	Line No.		Piping Class		Line Thickness	3"-SW-1424219-1BW3-STW	1BW3		S-40
PROCESS CONDITIONS	7	Fluid		Phase			Plant Water	Liquid		
	8	Flow:	Minimum	Normal	Maximum	Unit	-	-	9970.00	kg/h
	9	Temperature	Minimum	Normal	Maximum	Unit	-	40.0	-	°C
	10	Pressure:	Minimum	Normal	Maximum	Unit	-	3.4	-	bar g
	11	Base Temperature		Design Temperature		Unit	-	65		°C
	12	Base Pressure		Design Pressure		Unit	-	10.3		bar g
	13	Full Scale Flow				Unit	10967			kg/h
	14	Maximum allowable pressure loss				Unit	0.2			bar
	15	Operating Density		Density @ Base		Unit	996.1			kg/m3
	16	Compressibility Factor (Z)					3.056e-003			
	17	Molecular Weight		Cp / Cv			18.02	1.156		
	18	Vapor Pressure Pv				Unit	7.294e-002			bar a
	19	Maximum Velocity				Unit	0.6			m/s
	20	Vaccum Possibility		NACE Requirement			No	No		
	21	Operating Viscosity				Unit	0.6514			cp
	22	Corrosive Service		Cold Service			No	No		
ORIFICE PLATE	23	Orifice Plate Type					square-edged concentric			
	24	Standard					ISO 5167			
	25	Material					SS 316			
	26	Plate Thickness					3 mm			
	27	Pipe inside diameter (20°C)					3.068 in			
	28	Pipe Reynolds No. @ Operating Flow					76393.608061			
	29	Beta Ratio @ 20°C					0.45497395563			
	30	Bore Dia @ 20°C					1.395 in			
	31	Drain or Vent Hole Size		Hole Position			N/A (Note *)	N/A (Note *)		
	32	Residual pressure loss					96.84			mbar
	33	Sound pressure level					24.60			dB(A)
	34	Ring Material & Type					N/A			
	35	Taps Type	Taps Size	Taps Orientation			Flange	1/2" NPT	45° Down ward Horizontal	
	36	Flange Size		Flange Type & Rating			3"	RF & 300#		
	37	Flange Material		Gasket Material			A105	SPW,SS 316L Winding & Inner Ring,CS Outer Ring,Graphite Filler		
	38	Permanent Pressure Loss					96.84 mbar			
	39	Spacer Ring					N/A			
METER	40	Transmitter Spec.					Model : 2051C D 2 A 0 2 A 1 A S5 B4 I1 M5 Q4 Q8			
	41	Differential Pressure @ Full scale					125 mbar			
	42	Meter Max Flow					0 ~ 10967 Kg/h			
	43									
	44									
PURCHASE	45	Manufacturer					ARAMAK / IRAN			
	46	Model					Orifice Plate Primary :OFT-RF1-80-I1-A3-A2-I4-DH3-HA1-0-01-C0-DW-SG2 - GS-T02			
	47	Requisition Number		Item Number			-	-		
	48	Serial Number					N/A			

Figure A.2: Approved engineering datasheet for Plant Water Tag 42-FE-4272

```
ENTER D:
77.92
ENTER phase (Liquid/Gas):
gas
ENTER qv:
3300
ENTER P normal:
6.9
ENTER rho:
8.618
ENTER mu:
0.0199
ENTER kappa (Cp/Cv):
1.421
ENTER tap_type:
flange
ENTER delta p (mbar):
1000

qm_kg_h: 3300.000
d(mm): 38.46
d(inch): 1.514
beta: 0.4936
Re: 752696
C: 0.6031
epsilon: 0.9665
deltaP(mbar): 1000
pressure_loss(mbar): 739.45
uncertainty: 0.50
Total Iterations: 7
```

```
ENTER D:
77.92
ENTER phase (Liquid/Gas):
liquid
ENTER qv:
10967
ENTER P normal:
3.4
ENTER rho:
996.1
ENTER mu:
0.6514
ENTER kappa (Cp/Cv):
1.156
ENTER tap_type:
flange
ENTER delta p (mbar):
125

qm_kg_h: 10967.000
d(mm): 35.46
d(inch): 1.396
beta: 0.4551
Re: 76419
C: 0.6047
epsilon: 1.0000
deltaP(mbar): 125
pressure_loss(mbar): 96.83
uncertainty: 0.50
Total Iterations: 7
```

*Figure B.1 & B.2: C++ terminal execution output for Plant Air & Plant **Water***